

Unit : 1 : Foundation Agile & Devops

Modern software

- Modern software development required fast delivery better quality & continuous improvement.
- Traditional software development approaches often suffer from long development cycle, poor communication, delayed feedback & higher failure rates. to address these challenges to major approaches, emerge.

1. Agile:- It focuses on iterative software development & customer collaboration.

2. DevOps:- It focuses on collaboration between development and operation teams to automate and accelerate software delivery.

What is Agile?

Agile is software development methodology that include continuous delivery of software, customer collaboration, adaptation of changing requirements and incremental iterative development.

instead of building a system at once agile develops software in small increments for sprints.

Agile manifesto :-

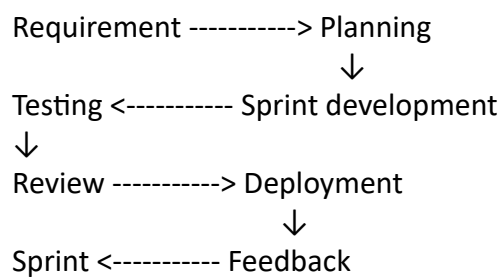
1. Individual and interactions over Process and tools.
2. working software over comprehensive documentary.
3. customer collaboration over contract negotiation.
4. Responding to change over following the Plan.

*** Agile Principles :-**

1. Customer satisfaction through early & continuous delivery.
2. Welcomes changing requirements.

3. Deliver working software frequently.
4. Business People & developers work together daily.
5. Build Projects around motivated individuals.
6. Face to Face Communication is most effective.
7. Working Software is the Primary measure of Progress.
- 8.
9. Continuous attention to the technical excellence.
10. Simplicity is essential.
11. Self-organizing team creates the best Solutions.
12. Teams regularly reflect & improves.

Agile lifecycle :-



Lean Thinking :-

Lean thinking originated from Together Production system (TPS)
It's goal is to deliver maximum Customer value with minimum waste.
Lean focuses on improving efficiency & eliminating activities that don't add values.

Lean Principles :-

- > Identify values & understand what Customers need.
- > map the value stream by identifying all steps required.
- > Create a flow which ensure throughout smooth execution.
- > Establish a bridge by producing only when needed.
- > Peruse perfection by continuously improving the process.

DevOps :-

- What is Devops ?
 - Devops is a combination of development + operations, cultural set of Practices and tools that enables orgination to delivery software rapidly and reliably.

The three ways of Devops

- (1) The flow :- It focuses on development, testing, deployment & then Customer
-> The main objective is to reduce the error, automate the work flows & faster delivery
Ex :- CI/CD Pipeline automatically deploys software
- (2) Feedback :- It's creates rapid feedback tools which contains operations then development & then improvement.
-> It's benefits by fixing the bug faster and improve reliability.
Ex :- monitoring tools instantly report failures
- (3) Continue learning :- It encourages experimentation, innovation and continuous learning.
Ex :- Conducting trials after every sprint.

Q : Find out devops benifits and failure modes ? Also search about cultural and team responsibility ?

1. Benefits of DevOps

DevOps is a set of practices that combines software development (Dev) and IT operations (Ops) to improve collaboration and productivity.

Benefits:

- 1. Faster Software Delivery**
 - Enables continuous integration and continuous deployment (CI/CD).
 - Reduces development and release time.
- 2. Improved Collaboration**
 - Developers and operations teams work together.
 - Better communication and teamwork.
- 3. Higher Quality Software**
 - Automated testing helps detect bugs early.

- Reduces errors in production.

4. Increased Reliability

- Continuous monitoring improves system stability.
- Quick identification and resolution of issues.

5. Greater Efficiency

- Automation reduces manual work.
- Saves time and resources.

6. Faster Recovery

- Quick rollback and recovery mechanisms.
- Minimizes downtime during failures.

7. Customer Satisfaction

- Faster delivery of features and updates.
- Better user experience.

2. DevOps Failure Modes

Failure modes are common reasons why DevOps implementations fail.

1. Lack of Cultural Change

- Teams continue working in silos.
- Poor communication between development and operations.

2. Resistance to Change

- Employees may hesitate to adopt new tools and processes.
- Slows DevOps adoption.

3. Insufficient Automation

- Manual processes increase errors and delays.
- Limits DevOps effectiveness.

4. Poor Monitoring and Feedback

- Problems remain undetected.
- Slower incident response.

5. Lack of Skills and Training

- Team members may not understand DevOps tools and practices.
- Leads to implementation issues.

6. Security Neglect

- Ignoring security during development and deployment.
- Can result in vulnerabilities and cyberattacks.

7. Unrealistic Expectations

- Expecting immediate results from DevOps adoption.
- Causes disappointment and project failure.

3. Cultural Aspects of DevOps

DevOps culture focuses on collaboration, communication, and continuous improvement.

Key Cultural Principles:

1. **Collaboration**
 - Development, Operations, and Security teams work together.
2. **Shared Responsibility**
 - Everyone is responsible for application quality and performance.
3. **Continuous Learning**
 - Teams regularly learn new technologies and practices.
4. **Transparency**
 - Open communication and information sharing.
5. **Customer-Centric Approach**
 - Focus on delivering value to customers.
6. **Continuous Improvement**
 - Regularly analyze processes and improve them.
7. **Trust and Accountability**
 - Team members trust each other and take ownership of their work.

4. Team Responsibilities in DevOps

Development Team

- Write and maintain application code.
- Perform unit testing.
- Collaborate with operations teams.

Operations Team

- Manage infrastructure and servers.
- Monitor system performance.

- Ensure application availability.

Quality Assurance (QA) Team

- Perform automated and manual testing.
- Verify software quality.

Security Team

- Identify and fix security risks.
- Implement security best practices.

DevOps Engineers

- Build CI/CD pipelines.
- Automate deployment and infrastructure.
- Integrate development and operations activities.

Product Owners/Managers

- Define business requirements.
- Prioritize features and releases.